Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$f(x) = \begin{cases} -x^2 & , \quad x \leq 0 \\ 2 & , x > 0 \end{cases}$, then

(A) $f(x)$ is a periodic function

(B) $f(x)$ is an odd function

(C) $f(x)$ is not differentiable at $x = 0$

(D) $f(x)$ is not continuous at $x = 1$

Q.2

Let $f : R \to R$ be a function defined by $f(x) = \max \cdot \{x, x^3\}$. The set of all points where $f(x)$ is not differentiable is

(A) $\{-1, 1\}$

(B) $\{-1, 1, 0\}$

(C) $\{0, 1\}$

(D) $\{-1, 0, 1\}$

Q.3

$f(x) = 1 + [\cos x]x$, in $0 < x \leq \frac{\pi}{2}$ where [.] is GIF, then $f(x)$ is:

(A) is continuous in $\left[0, \frac{\pi}{2}\right]$

(B) has a maximum value 2

(C) has a minimum value 0

(D) is not differentiable at $x = \pi/2$

Q.4

If $f(2) = g(2)$ and $f'(2) = a, g'(2) = 2a$, then $\lim_{x \to 2} \frac{f(x) - g(x)}{x - 2}$ is equal to

(A) 1

(B) a

(C) $-a$

(D) none of these

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Q.5

Let $f'(x) = 3x^2 \cdot \sin \frac{1}{x} - x \cos \frac{1}{x}, x \neq 0, f(0) = 0, f\left(\frac{1}{\pi}\right) = 0$ then which of the following is correct.

(A) $f(x)$ is continuous at $x = 0$

(B) $f(x)$ is non-differentiable at $x = 0$

(C) $f'(x)$ is discontinuous at $x = 0$

(D) $f'(x)$ is differentiable at $x = 0$

Q.6

If $f(x)$ is continuous and $f\left(\frac{9}{2}\right) = \frac{2}{9}$, then $\lim_{x \to 0} f\left(\frac{1-\cos 3x}{x^2}\right)$ is equal to

(A) $\frac{9}{2}$

(B) $\frac{2}{9}$

(C) 0

(D) none of these

Q.7

Let $f(x) = \lim_{n \to \infty} \frac{x}{1+(2 \sin x)^{2n}}$, then f is discontinuous at

(A) π

(B) $\pi/3$

(C) $\pi/4$

(D) $\pi/6$

Q.8

At which points the function $f(x) = \frac{x}{[x]}$, where [.] is greatest integer function, is discontinuous

(A) only positive integers

(B) all integer and $(0, 1)$

(C) all rational numbers

(D) none of these

Q.9

If $f(x) = x, x \leq 1$ and $f(x) = x^2 + bx + c, x > 1$ and $f'(x)$ exists finitely for all $x \in R$, then

(A) $b = -1, c \in R$

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(B) $c = 1, b \in R$

(C) $b = 1, c = -1$

(D) $b = -1, c = 1$

Q.10

If $f(x) = (\sin x)^{\frac{1}{\pi - 2x}}, x \neq \frac{\pi}{2}$. If $f(x)$ is continuous at $x = \frac{\pi}{2}$, then $f\left(\frac{\pi}{2}\right)$ is

(A) e

(B) 1

(C) 0

(D) none of these

Q.11

If $f(x)$ is defined by $f(x) = \begin{cases} \frac{|x^2 - 2x|}{x^2 - 2x} & x \neq 0, 2 \\ 1 & x = 0 \\ -1 & x = 2 \end{cases}$ then f is discontinuous at

(A) -1 and 1

(B) 0 and 2

(C) 0 and 1

(D) -2 and 2

Q.12

Number of points of discontinuity of $f(x) = \frac{1}{u^2 + u - 2}, \quad u = \frac{1}{x - 1}$ is

(A) 1

(B) 2

(C) 3

(D) 4

Q.13

If $f(x) = \begin{cases} ax^2 + b, & x \leq 0 \\ 0, & x > 0 \end{cases}$ is differentiable at $x = 0$, then

(A) $a = 1, b = 1$

(B) $a = 1, b = 2$

(C) $a = 2, b = 0$

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(D) $a = 2, b = 1$

Q.14

The set of points where the function $f(x) = x|x|$ is differentiable is

(A) $(-\infty, \infty)$

(B) $(-\infty, 0) \cup (0, \infty)$

(C) $(0, \infty)$

(D) $[0, \infty]$

Q.15

The value of $f(0)$ so that the function $f(x) = \frac{2x - \sin^{-1} x}{2x + \tan^{-1} x}$ is continuous at each point on its domain is

(A) 2

(B) $\frac{1}{3}$

(C) $\frac{2}{3}$

(D) $-\frac{1}{3}$

Q.16

If $x + 2|y| = 3y$, where $y = f(x)$, then $f(x)$ is

(A) continuous everywhere

(B) differentiable everywhere

(C) discontinuous at $x = 0$

(D) none of the above

Q.17

Let $f(x) = \sin x, g(x) = [x + 1]$ and $g\{f(x)\} = h(x)$, where [.] is the greatest integer $h'\left(\frac{\pi}{2}\right)$ function. Then

(A) non existent

(B) 1

(C) -1

(D) none of these

Q.18

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Let f be the function defined by $f(x) = \begin{cases} \frac{x^2-1}{x^2-2|x-1|-1} & ; \quad x \neq 1 \\ 1/2 & ; \quad x = 1 \end{cases}$

(A) the function is continuous for all x

(B) the function is continuous only for $x > 1$

(C) the function is continuous at $x = 1$

(D) the function is not continuous at $x = 1$

Q.19

$f(x) = \frac{x}{1+[x]}$, is discontinuous in set s, where s is

(A) $\{[-1, 0) \cup$ integers$\}$

(B) \{all real numbers\}

(C) \{all positive rational\}

(D) none of these

Q.20

The function $f(x) = \max\{(1-x), (1+x), 2\}$ is where $x \in (-\infty, \infty)$

(A) discontinuous at all points

(B) differentiable at all points

(C) differentiable at all points except - 1 and 1

(D) continuous at all points except -1 and 1

Q.21

Let $f(x) = [4 + 3\cos x], x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ where $[x]$ = greatest integer less than or equal to x. The number of points of discontinuity of $f(x)$ is

Q.22

The value of $f(0)$, so that the function $f(x) = \frac{(27-2x)^{1/3}-3}{9-3(243+5x)^{1/5}}, (x \neq 0)$ is continuous, is given by

Q.23

If for a differentiable function $f, f(0) = f(1) = 0, f'(1) = 2$ and $g(x) = f\left(e^x\right) e^{f(x)}$, then $g'(0)$ is

Q.24

If $f(x) = p|\sin x| + qe^{|x|} + r|x|^3$ and if $f(x)$ is differentiable at $x = 0$, then $p + q =$

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Q.25

If the function $f(x)=\begin{cases}\frac{3}{x^2}\sin 2x^2 & ; \quad \text{if } x<0 \\ 0 & ; \quad \text{if } x=2 \\ \frac{x^2-3x+k}{1-3x^2} & ; \quad \text{if} \quad x\geq 0, x\neq 2\end{cases}$ is continuous at $x=0$, then the value of k is

Q.26

Let $f(2)=4$ and $f'(2)=4$. Then $\lim_{x\to 2}\frac{xf(2)-2f(x)}{x-2}$ is given by L. Then $100+L=$

Q.27

If $f(x)=\begin{cases}\sin x, & x\neq n\pi, n\in Z \\ 0, & \text{otherwise}\end{cases}$ and $g(x)=\begin{cases}x^2+1, x\neq 0,2 \\ 4, \quad x=0 \\ 5, \quad x=2\end{cases}$, then $\lim_{x\to 0} g\{f(x)\}$ is equal to

Q.28

The value of $f(0)$, so that the function $f(x)=\frac{\sqrt{a^2-ax+x^2}-\sqrt{a^2+ax+x^2}}{\sqrt{a+x}-\sqrt{a-x}}$ becomes continuous for all x, is given by $-a^{m/n}$ where m and n are coprime. Then $100mn=$

Q.29

Let $f(x)=x^3-x^2+x+1, \quad g(x)=\begin{cases}\max\cdot\{f(t), 0\leq t\leq x\}, & 0\leq x\leq 1 \\ 3-x, & 1<x\leq 2\end{cases}$
Then in $[0,2]$ the number of points where $g(x)$ is not differentiable is

Q.30

f and g are two differentiable function which satisfy the condition $g'(a)=2, g(a)=b$ and $(fog)=I$ (identity function), then the reciprocal of $f'(b)$ is equal to

Answers & Solutions

Q.1 Answer:

$f(x)$ is not differentiable at $x=0$

Solution:

$f(x)$ **is not continuous at** $x=0$ **(LHL** $\neq$ **RHL), so not differentiable at** $x=0$**.**

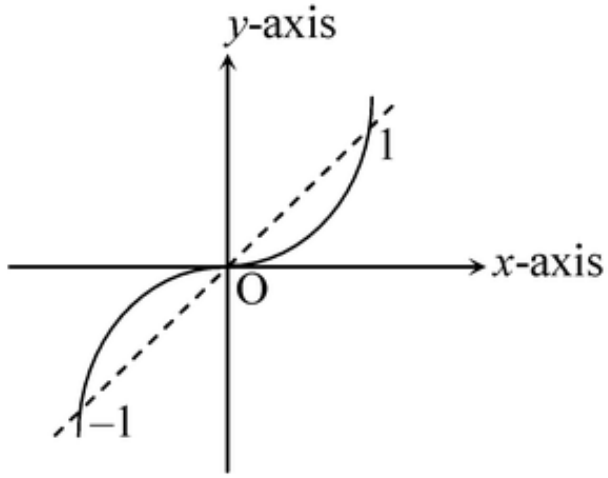
Q.2 Answer:

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$\{-1, 0, 1\}$

Solution:

$f(x) = \max \cdot \{x, x^3\}$. **From the graph, itself the function is not differentiable at** $\{-1, 0, 1\}$ **due to the nature of the graph.**



Q.3 Answer:

is continuous in $\left[0, \frac{\pi}{2}\right]$

Solution:

$$f(x) = 1 + [\cos x]x \text{ in } 0 < x \leq \pi/2$$
$$\text{In } 0 < x \leq \pi/2, \cos x \in [0, 1)$$
$$\Rightarrow f(x) = 1 + 0.x = 1$$

Since $f(x)$ **is a constant function, it is continuous as well as differentiable in the entire interval.**

Q.4 Answer:

$-a$

Solution:

$$\lim_{x\to 2} \frac{f(x)-g(x)}{x-2} = \lim_{x\to 2} \frac{f'(x)-g'(x)}{1} = f'(2) - g'(2) = a - 2a = -a$$

Q.5 Answer:

$f(x)$ is continuous at $x = 0$

Solution:

$$f'(x) = 3x^2 \cdot \sin\frac{1}{x} - x \cdot \cos\frac{1}{x}$$
$$\Rightarrow f(x) = \int \left(3x^2 \cdot \sin\frac{1}{x} - x\cos\frac{1}{x}\right) dx = \sin\frac{1}{x} \cdot x^3 - \int \cos\frac{1}{x}\left(-\frac{1}{x^2}\right) x^3 dx -$$
$$\int x\cos\frac{1}{x} dx$$
$$= x^3 \sin\frac{1}{x} + c \quad , \quad \text{since} \quad f\left(\frac{1}{\pi}\right) = 0 + c$$
$$\Rightarrow f(x) = \begin{cases} x^3 \sin\frac{1}{x}, & x \neq 0 \\ 0 \quad , & x = 0 \end{cases}$$

$f(x)$ **is clearly continuous and differentiable at** $x = 0$ **zero with** $f'(0) = 0$.

$$f''(0) = \lim_{h\to 0} \frac{3h^2 \sin\frac{1}{h} - h\cos\frac{1}{h}}{h} = 3h \sin\frac{1}{h} - \cos\frac{1}{h}$$

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This limit doesn't exist, hence $f'(x)$ is non-differentiable at $x=0$ also

$\lim_{x\to 0} f'(x) = 0$

Thus $f'(x)$ is continuous at $x=0$.

Q.6 Answer:

$\frac{2}{9}$

Solution:

$\lim_{x\to 0} f\left(\frac{1-\cos 3x}{x^2}\right) = f\left(\lim_{x\to 0}\frac{1-\cos 3x}{x^2}\right)$ **(since f is continous)**

$f\left(\lim_{x\to 0}\frac{2\sin^2\frac{3x}{2}}{x^2}\right) = f\left(\lim_{x\to 0} 2\cdot\left(\frac{\sin\frac{3x}{2}}{3x/2}\right)^2\cdot\frac{9}{4}\right) = f\left(\frac{9}{2}\right) = \frac{2}{9}$

Hence $\lim_{x\to 0} f\left(\frac{1-\cos 3x}{x^2}\right) = \frac{2}{9}$

Q.7 Answer:

$\pi/6$

Solution:

$$f(x) = \begin{cases} x & \text{if} \quad |\sin x| < \frac{1}{2} \text{ i.e. } -\frac{\pi}{6}+n\pi < x < \frac{\pi}{6}+n\pi \\ \frac{x}{2} & \text{if} \quad |\sin x| = \frac{1}{2} \text{ i.e. } x = \pm\frac{\pi}{6}+n\pi \\ 0 & \text{if} \quad |\sin x| > \frac{1}{2} \text{ i.e. } \frac{\pi}{6}+n\pi < x < \frac{5\pi}{6}+n\pi \end{cases}$$

Thus points of discontinuities of f are of the form $x = \pm\frac{\pi}{6}+n\pi$ (LHL $\neq$ RHL)

In particular, f is discontinuous at $x=\frac{\pi}{6}$

Q.8 Answer:

all integer and $(0,1)$

Solution:

(i) When $0\le x<1$ $f(x)$ doesn't exist as $[x]=0$ here. $\lim_{x\to l+} f(x)$ and $\lim_{x\to l-} f(x)$ does not exist.

(ii) Also Hence $f(x)$ is discontinuous at all integers and also in $(0,1)$

Q.9 Answer:

$b=-1, c=1$

Solution:

$f(x)$ is differentiable at $x=1$

$\therefore \lim_{h\to 0}\frac{f(1+h)-f(1)}{h} = \lim_{h\to 0}\frac{f(1-h)-f(1)}{-h}$

Now $\lim_{h\to 0}\frac{f(1+h)-f(1)}{h} = \lim_{h\to 0}\frac{(1+h)^2+b(1+h)+c-1}{h}$

$= \lim_{h\to 0}\frac{h^2+(2+b)h+b+c}{h}$

$\lim_{h\to 0}\frac{f(1-h)-f(1)}{-h} = \lim_{h\to 0}\frac{1-h-1}{-h} = 1$

The two limits can be equal if $2+b=1, b+c=0$

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Q.10 Answer:

1

Solution:

Here, $\lim_{h\to 0}\left\{\sin\left(\frac{\pi}{2}+h\right)\right\}^{\frac{1}{\pi-2(\pi/2+h)}}=\lim_{h\to 0}\left\{\sin\left(\frac{\pi}{2}-h\right)\right\}^{\frac{1}{\pi-2(\pi/2-h)}}=f\left(\frac{\pi}{2}\right)$
Now, $\lim_{h\to 0}(\cos h)(-2h)=e^{\lim(1/(-2h))\log\cos h}=e^{\lim_{h\to 0}\frac{(1/\cos h)(-\sin h)}{-2}}=e^0=1$
$\therefore f\left(\frac{\pi}{2}\right)=1$

Q.11 Answer:

0 and 2

Solution:

$f(x)=\begin{cases}1 & x\le 0\\ -1 & 0<x\le 2\\ 1 & x>2\end{cases}$ **. Clearly** f **is discontinuous at 0 and 2 only.**

Q.12 Answer:

3

Solution:

The function $\mathrm{u}=\frac{1}{\mathrm{x}-1}$ **suffers a discontinuity at the point** $\mathrm{x}=1$ **The function** $\mathrm{f(x)}=\frac{1}{\mathrm{u}^2+\mathrm{u}-2}$ **suffers a discontinuity at the points where** $\mathrm{u}^2+\mathrm{u}-2=0$ **i.e.,** $\mathrm{u}=-2$ **and** $\mathrm{u}=1$.
Using these value of u**, the corresponding values of** x **are obtained by solving the equations** $-2=\frac{1}{x-1}$ **and** $1=\frac{1}{x-1}$ **i.e.,** $x=\frac{1}{2}$ **and** $x=2$
Hence, the composite function is discontinuous at three points $\mathrm{x}=\frac{1}{2},\mathrm{x}=1$ **and** $x=2$.

Q.13 Answer:

$a=2,b=0$

Solution:

For continuity at $x=0,b=0$
For differentiability $2ax=0$ **at** $x=0$
So a **can be any real number**

Q.14 Answer:

$(-\infty,\infty)$

Solution:

$f(x)=-x^2,x<0$
$=x^2,\quad x\ge 0$
The function is differentiable for all $x>0$ **and for all** $x<0$.

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Take differentiability for $x = 0$. S.H.D. $\lim_{h\to 0} \frac{-(-h)^2-0}{-h} = 0$

R.H.D. $= \lim_{h\to 0} \frac{h^2-0}{h} = 0$

Q.15 Answer:

$\frac{1}{3}$

Solution:

$\lim_{x\to 0} f(x) = f(0) \quad \Rightarrow \quad \lim_{x\to 0} \frac{2x-\sin^{-1} x}{2x+\tan^{-1} x}$

$= \lim_{x\to 0} \frac{2-\frac{\sin^{-1} x}{x}}{2+\frac{\tan^{-1} x}{x}} = \frac{2-1}{2+1} = \frac{1}{3}; \quad \therefore f(0) = \frac{1}{3}$

Q.16 Answer:

continuous everywhere

Solution:

For $\mathrm{f(x)} \geq 0, |\mathrm{f(x)}| = \mathrm{f(x)}.$

So,

$\mathrm{x} + 2|\mathrm{f(x)}| = 3\mathrm{f(x)}$

$\Rightarrow \mathrm{x} + 2\mathrm{f(x)} = 3\mathrm{f(x)}$

$\Rightarrow \mathrm{f(x)} = \mathrm{x}$

But since this is for $\mathrm{f(x)} \geq 0, \mathrm{f(x)} = \mathrm{x} \quad \forall \mathrm{x} \geq 0.$

For $\mathrm{f(x)} < 0, |\mathrm{f(x)}| = -\mathrm{f(x)}$

So, $\mathrm{x} + 2|\mathrm{f(x)}| = 3\mathrm{f(x)}$

$\Rightarrow x - 2f(x) = 3f(x)$

$\Rightarrow f(x) = \frac{x}{5}$

But since this is for $\mathrm{f(x)} < 0, \mathrm{f(x)} = \frac{\mathrm{x}}{5} \quad \forall \mathrm{x} < 0$

Checking for continuity at $\mathrm{x} = 0$ **:**

$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{x}{5} = 0$

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x = 0$

$f(0) = 0$

Since it is an algebraic function that changes its definition only at $\mathrm{x} = 0$**, the function is continuous everywhere.**

Checking for differentiability at $\mathrm{x} = 0$ **: the slope of the function changes at this point, so it is not differentiable here.**

So, option A is the correct option

Q.17 Answer:

non existent

Solution:

$h(x) = g\{f(x)\} = [f(x) + 1] = [\sin x + 1]$

$h'\left(\frac{\pi}{2} + 0\right) = \lim_{h\to 0} \frac{\left[\sin\left(\frac{\pi}{2}+h\right)+1\right]-\left[\sin\frac{\pi}{2}+1\right]}{h} = \lim_{h\to 0} \frac{[\cos h+1]-[2]}{h}$

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$= \lim_{h\to 0} \frac{1-2}{h} = \lim_{h\to 0} \frac{-1}{h} = -\infty$

$h'\left(\frac{\pi}{2}-0\right) = \lim_{h\to 0} \frac{\left[\sin\left(\frac{\pi}{2}-h\right)+1\right]-\left[\sin\frac{\pi}{2}+1\right]}{-h} = \lim_{h\to 0} \frac{[\cos h+1]-[2]}{-h} = \lim_{h\to 0} \frac{1}{h} = \infty$

Q.18 Answer:

the function is not continuous at $x = 1$

Solution:

$x < 1, f(x) = \frac{x^2-1}{x^2+2x-3} = \frac{x+1}{x+3}; \quad \therefore \quad \lim_{x\to 1^-} f(x) = \frac{1}{2}$

For $x > 1, f(x) = \frac{x^2-1}{x^2-2x+1} = \frac{x+1}{x-1}$

$\therefore \lim_{x\to 1^+} f(x) = \infty$

$\therefore$ **The function is not continuous at** $x = 1$

Q.19 Answer:

$\{[-1, 0) \cup$ integers$\}$

Solution:

$f(x) = \frac{x}{1+n}, \quad n \le x < n+1$

(i) clearly, if $n = -1$ **then** $f(x)$ **is not defined**

i.e. $f(x)$ **is not defined** $\forall x \in [-1, 0)$

(ii) if $n \neq -1$

$$f(x) = -\frac{x}{2}; \qquad -3 \le x < -2 \quad (n = -3)$$
$$= -x; \qquad -2 \le x < -1 \quad (n = -2)$$
$$= x; \qquad 0 \le x < +1 \qquad (n = 0)$$
$$= \ldots\ldots\ldots$$

Clearly, the function is discontinuous at all integral values of x**.**

From (i) and (ii) $f(x)$ **is not continuous in set 'S'**

$S = \{[-1, 0) \cup \text{ integers }\}.$

Q.20 Answer:

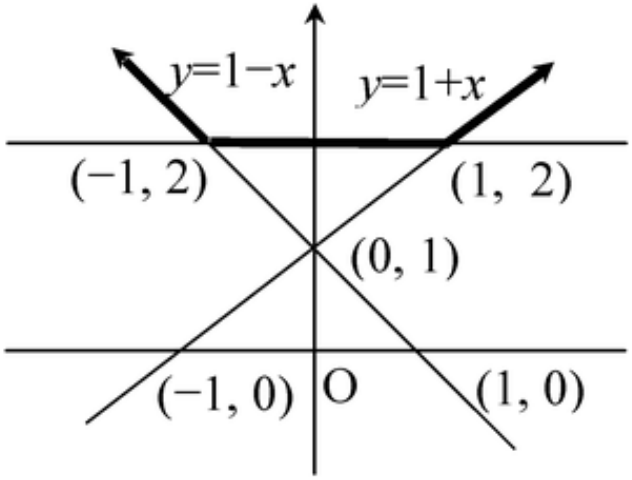
differentiable at all points except - 1 and 1

Solution:

We draw the graph of $y = 1 - x, y = 1 + x$ **and** $y = 2\, f(x) = \max \cdot \{1 - x, 1 + x, 2\}$

$$f(x) = \begin{cases} 1-x, & x \le -1 \\ 2, & -1 < x < 1 \\ 1+x, & x \ge 1 \end{cases}$$

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From graph it is clear that $f(x)$ is continuous at all x and differentiable at all x except $x = -1$ and $x = 1$

Q.21 Answer:

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Solution:

$4 < 4 + 3\cos x \leq 7$ **for** $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$f(x) = [4 + 3\cos x]$ **is discontinuous at those points where** $4 + 3\cos x$ **is an integer**

Now $4 + 3\cos x = 4$ **if** $\cos x = 0$**. So** $x = \frac{-\pi}{2}, \frac{\pi}{2}$ **(not possible)**

$4 + 3\cos x = 5$ **if** $\cos x = \frac{1}{3}$**. So** x **has two values**

$\cos^{-1}\left(\frac{1}{3}\right)$ **and** $-\left(\cos^{-1}\frac{1}{3}\right)$

$\cos^{-1}\frac{2}{3}, -\cos^{-1}\frac{2}{3}$

$4 + 3\cos x = 6$ **if** $\cos x = \frac{2}{3}$**. So** x **has two values**

$4 + 3\cos x = 7$ **if** $\cos x = 1$**. So** $x = 0$

$\therefore$ **the number of values of** $x = 2 + 2 + 1 = 5$

Q.22 Answer:

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Solution:

Since $f(x)$ **s continuous at** $x = 0$**, therefore**

$$f(0) = \lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{(27-2x)^{1/3} - 3}{9 - 3(243+5x)^{1/5}} \quad \left(\text{form } \frac{0}{0}\right)$$

$$\lim_{x\to 0} \frac{\frac{1}{3}(27-2x)^{-2/3}(-2)}{-\frac{3}{5}(243+5x)^{-4/5}(5)}$$

$$= \frac{1}{5} = 2$$

Q.23 Answer:

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Solution:

$g(x) = f\left(e^x\right) e^{f(x)}$
$\therefore g'(x) = f'\left(e^x\right) \cdot e^x \cdot e^{f(x)} + f\left(e^x\right) \cdot e^{f(x)} \cdot f'(x)$
Put $x = 0$ **and** $f(0) = f(1) = 0, f'(1) = 2$
$\therefore \quad g'(0) = 2.1.1 + 0 = 2$

Q.24 Answer:

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Solution:

For $-\frac{\pi}{2} < x \leq 0, f(x) = -p \sin x + qe^{-x} - rx^3$
So, $f'(0-) = \lim_{x \to 0-} \frac{f(x)-f(0)}{x-0} = \lim_{x \to 0-} \left[-\frac{p \sin x}{x} - q\left(\frac{e^{-x}-1}{-x}\right) - rx^2\right] = -p - q$
For $0 < x < \frac{\pi}{2}, f(x) = p \sin x + qe^x + rx^3$
$f'(0+) = \lim_{x \to 0+} \frac{f(x)-f(0)}{x-0} = \lim_{x \to 0+} \left[\frac{p \sin x}{x} + q\left(\frac{e^x-1}{x}\right) - rx^2\right] = p + q$
For f **to be differentiable at** $x = 0$**, we must have**
$p + q = -p - q \quad \Rightarrow \quad p + q = 0$

Q.25 Answer:

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Solution:

$$\lim_{x \to 0-} f(x) = \lim_{x \to 0} \frac{3}{x^2} \sin\left(2x^2\right)$$
$$= \lim_{x \to 0} \frac{6}{2x^2} \sin\left(2x^2\right) = 6$$
$$\lim_{x \to 0+} f(x) = \lim_{x \to 0} \frac{x^2 - 3x + k}{1 - 3x^2} = k$$

Since $f(x)$ **is continuous at** $x = 0 \therefore \quad \lim_{x \to 0-} f(x) = \lim_{x \to 0+} f(x) = f(0)$
$\therefore \quad k = 6$

Q.26 Answer:

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Solution:

$\lim_{x \to 2} \frac{xf(2) - 2f(x)}{x-2} = \lim_{x \to 2} \frac{f(2) - 2f'(x)}{1} = f(2) - 2f'(2) = 4 - 2.4 = -4$

Q.27 Answer:

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Solution:

$\lim_{x \to 0} g(f(x)) = \lim_{x \to 0} [f(x)]^2 + 1 = \lim_{x \to 0} \left(\sin^2 x + 1\right) = 1$

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Q.28 Answer:

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Solution:

As $f(x)$ is continuous at $x=0$, therefore $\lim_{x\to 0} f(x) = f(0)$

$\Rightarrow f(0) = \lim_{x\to 0} \frac{\sqrt{a^2-ax+x^2}-\sqrt{a^2+ax+x^2}}{\sqrt{a+x}-\sqrt{a-x}}$

$\Rightarrow f(0) = \lim_{x\to 0} \frac{-2ax(\sqrt{a+x}+\sqrt{a-x})}{\left(\sqrt{a^2-ax+x^2}+\sqrt{a^2+ax+x^2}\right)2x} = \frac{-a2\sqrt{a}}{2a} = -\sqrt{a}$

Q.29 Answer:

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Solution:

$f(t) = t^3 - t^2 + t + 1$

$f'(t) = 3t^2 - 2t + 1 > 0$

$\therefore$ $f(t)$ is an increasing function.

Since $0 \le t \le x$

$\therefore$ max. $f(t) = f(x) = x^3 - x^2 + x + 1$

Thus $g(x) = x^3 - x^2 + x + 1, 0 \le x \le 1$ and $g(x) = 3 - x, 1 < x \le 2$

The only doubtful point for differentiability of $g(x)$ in $[0, 2]$ is $x = 1$.

Clearly, $\lim_{x\to 1^+0} g(x) = \lim_{x\to 1^+}(3-x) = 2$ and $g(1) = 2$

$\therefore$ $g(x)$ is continuous at $x = 1$

Also $g'(x) = 3x^2 - 2x + 1, 0 \le x < 1$ and $g'(x) = -1, 1 < x \le 2$

$\therefore$ $g'(1-0) = 3\cdot 1^2 - 2\cdot 1 + 1 = 2$ and $g'(1+0) = -1$

Hence $g(x)$ is not differentiable at $x = 1$.

Q.30 Answer:

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Solution:

$(f \circ g)x = I(x) = x$

or $f[g(x)] = x$

Differentiating both sides $f'[g(x)]g'(x) = 1$

Put $x = a$, $f'[g(a)]g'(a) = 1$

Now, put $g(a) = b$ and $g'(a) = 2$

$f'[b]\cdot 2 = 1 \quad \Rightarrow \quad f'(b) = \frac{1}{2}$

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